UQA Reference Guide

1. Introduction

1.1 What is UQA?

UQA (Unified Query Algebra) is a database prototype that unifies four traditionally isolated data paradigms — relational SQL, full-text search, vector search, and graph queries — under a single algebraic structure, using **posting lists** as the universal abstraction.

Where conventional systems force users to stitch together a relational database, a search engine, a vector store, and a graph database — each with its own query language, data model, and operational overhead — UQA provides a single engine that speaks SQL enriched with cross-paradigm functions.

```sql
SELECT title, _score FROM papers
WHERE fuse_log_odds(
    text_match(title, 'attention'),
    knn_match(5),
    traverse_match(1, 'cited_by', 2)
) AND year >= 2020
ORDER BY _score DESC;
```

This single query combines BM25 text scoring, KNN vector similarity, and graph traversal reachability into a fused relevance score, then applies a relational filter on publication year — an operation that would require coordinating three separate systems in a conventional stack.

1.2 Design Principles

UQA is built on three principles:

1. **Algebraic unification.** Every paradigm — relational filtering, term-based retrieval, vector similarity, graph traversal — produces the same data structure: an ordered posting list. Boolean algebra (union, intersection, complement) operates uniformly across all of them.
2. **Probabilistic scoring.** Relevance signals are not raw scores but calibrated probabilities in $[0, 1]$. This enables principled multi-signal fusion through Bayesian inference rather than ad-hoc weighting.
3. **SQL as the interface.** Standard SQL is extended with domain-specific functions (`text_match`, `knn_match`, `traverse`, `fuse_log_odds`) rather than replaced by a novel query language. Users retain the full power of relational SQL — joins, subqueries, CTEs, window functions, transactions — while gaining access to cross-paradigm operations.

1.3 Project Status

UQA is a research prototype implementing the theoretical framework described in a series of four papers (see Section 2). It includes a complete SQL compiler, SQLite-backed persistence, a cost-based query optimizer, parallel execution, and disk spilling. It is designed as a proof-of-concept for the unified query algebra rather than a production database.

2. Theoretical Foundations

UQA is grounded in four papers that progressively build the mathematical framework.

2.1 Paper 1: A Unified Mathematical Framework for Query Algebras Across Heterogeneous Data Paradigms (December 2023)

This foundational paper establishes **posting lists** as a universal abstraction for result sets across relational, text retrieval, and vector search paradigms. A posting list is an ordered sequence of (document_id, payload) pairs, where payloads carry paradigm-specific metadata — field values for relational queries, term positions and scores for text retrieval, similarity scores for vector search.

Key results:

- **Boolean algebra.** The set of posting lists under union, intersection, and complement forms a complete Boolean algebra isomorphic to the power set of documents (Theorem 2.1.2). This means any combination of cross-paradigm queries can be composed using the same three operations.
- **Operator calculus.** Primitive operators — T (TermRetrieval), V_θ (VectorSimilarity), KNN_k, Filter, Score, Facet — are formally defined as functions producing posting lists. Their compositions form a monoid under function composition (Theorem 3.2.3), and derived operators for hybrid queries (HybridTextVector, SemanticFilter) are constructed from these primitives.
- **Join operations.** Cross-paradigm joins — TextVectorJoin, VectorRelationalJoin, and a generalized multi-paradigm join — are formally defined with proofs of commutativity (up to payload ordering) and associativity (Theorems 4.3.1, 4.3.2). Generalized posting lists carry multi-document tuples for join results.
- **Hierarchical data.** Path expressions and unnest operators handle nested JSON-like documents within the posting list framework.
- **Category theory.** A query category with posting lists as objects and operators as morphisms is established. Functors between paradigm-specific subcategories (relational, text, vector) formalize the cross-paradigm transformations. Natural transformations enable systematic rewriting rules for query optimization.
- **Query optimization.** Lattice-based rewriting rules — filter pushdown, vector threshold merge, intersect reordering by cardinality — are proven to preserve semantic equivalence while reducing computational cost.

2.2 Paper 2: Extending the Unified Mathematical Framework to Support Graph Data Structures (December 2024)

This paper extends the framework to incorporate graph data — vertices, edges, traversals, and pattern matching — as first-class citizens.

Key results:

- **Graph posting lists.** A graph posting list carries (id, subgraph) pairs. A bijective mapping between graph posting lists and standard posting lists establishes an isomorphism that

preserves all Boolean algebra operations (Theorem 1.1.6).

- **Graph operators.** Three fundamental graph operators are defined: **Traverse** (BFS from a starting vertex with label filtering and hop limit), **PatternMatch** (subgraph isomorphism with vertex/edge constraints), and **Regular Path Query** (NFA-based evaluation of path expressions like `cited_by/cited_by`).
- **Cross-paradigm integration.** Operators for converting between graphs and relational data (ToGraph, FromGraph), graph-vector integration (vertex embeddings, vector-enhanced pattern matching), and graph-text integration (semantic graph search) are formally defined.
- **Complexity analysis.** Pattern matching is NP-complete in the worst case ($O(|V_G|^{|V_P|})$); regular path queries are polynomial ($O(|V|^2 \cdot |R|)$); traversal is $O(\sum_{i=1}^{k} d^i)$ where d is the average degree.

2.3 Paper 3: Bayesian BM25 — A Probabilistic Framework for Hybrid Text and Vector Search (January 2026)

This paper addresses a five-decade gap in information retrieval: BM25 scores, despite being derived from a probabilistic model, are not probabilities. They are unbounded, query-dependent, and incompatible with other ranking signals. Bayesian BM25 closes this gap.

The problem. Standard BM25 scores range over $[0, +\infty)$. Directly combining them with bounded signals like cosine similarity ($[0, 1]$) causes signal dominance (Theorem 1.2.2). Reciprocal Rank Fusion (RRF) discards score magnitudes, losing confidence information (Theorem 1.3.2).

The solution. Bayesian BM25 applies Bayes' theorem with a **sigmoid likelihood model** ($P(s \mid R{=}1) = \sigma(\alpha(s - \beta))$), a **composite prior** incorporating term frequency and document length signals (Definition 4.2.3), and a **corpus-level base rate prior** capturing the global prevalence of relevance (Definition 4.4.1).

Key results:

- **Three-term posterior decomposition.** The log-odds of the posterior decomposes additively into likelihood, base rate, and document prior contributions (Theorem 4.4.2), enabling efficient computation via two successive Bayes updates.
- **Monotonicity preservation.** The Bayesian transformation preserves the ranking order of BM25 for fixed prior (Theorem 4.3.1), ensuring that higher BM25 scores still indicate higher relevance probability.
- **Calibration.** Expected calibration error is reduced by 68-77% compared to raw sigmoid calibration, without requiring relevance labels (Section 4.5).
- **WAND/BMW compatibility.** The transformation preserves safe document pruning through modified upper bounds that account for document-dependent priors (Theorem 6.1.2). Block-Max WAND partitions posting lists into fixed-size blocks and precomputes per-block maximum scores, enabling safe skipping of entire blocks during top-k retrieval.
- **Hybrid search fusion.** Calibrated probability signals from BM25 and vector similarity can be combined through probabilistic AND (product rule) and OR (inclusion-exclusion) operations. Log-space computation ensures numerical stability for extreme probabilities (Section 5.5).
- **Progressive parameter estimation.** The sigmoid parameters α and β can be estimated online from the score distribution without relevance labels, using median and IQR

heuristics (Section 8.3).

2.4 Paper 4: From Bayesian Inference to Neural Computation (February 2026)

This paper demonstrates that feedforward neural network structure **emerges analytically** from first-principles Bayesian inference over multiple relevance signals. Starting from the question "what is the probability that a document is relevant given multiple evidence signals?", the derivation arrives at a neural network without designing one.

The derivation proceeds in three stages:

1. **Calibration.** Each scoring signal produces a calibrated probability $P_i \in (0, 1)$.
2. **Log-odds aggregation.** Probabilities are mapped to log-odds via the logit function and combined as a weighted sum (normalized Logarithmic Opinion Pooling).
3. **Posterior computation.** The scaled mean log-odds passes through a sigmoid to produce the final probability.

Key results:

- **Neural structure as theorem.** The end-to-end computation has the structure of a feedforward neural network: inputs are calibrated to probabilities, pass through a logit nonlinearity (hidden layer), undergo linear aggregation with confidence scaling, and pass through a sigmoid output activation (Theorem 5.2.1). When all signals share sigmoid calibration, the logit-sigmoid identity collapses the hidden layer, yielding logistic regression (Theorem 5.2.1a). When signals have heterogeneous calibrations (the hybrid search case with BM25 + vector similarity), the logit performs a genuine nonlinear transformation, producing a true two-layer network.
- **Activation functions as probabilistic answers.** The sigmoid answers "how probable?" (Bayesian posterior for binary relevance). ReLU answers "how much?" (MAP estimator under sparse non-negative priors, Theorem 6.5.3). Swish answers "what is the expected relevant amount?" (Bayesian expected value under the same sparse gating structure, Theorem 6.7.4). GELU is the Gaussian approximation of Swish (Theorem 6.8.1). The ReLU-to-Swish transition is the MAP-to-Bayes duality of classical statistics applied to neural activations.
- **Log-odds conjunction resolves shrinkage.** Naive probabilistic conjunction (product rule) causes scores to shrink toward zero as more signals agree — violating the intuition that agreement should increase confidence. The log-odds mean with multiplicative confidence scaling ($\sqrt{n}$ law) preserves scale neutrality and sign: agreement among irrelevant signals cannot produce a relevance judgment (Corollary 4.2.3), a structural guarantee absent from additive formulations.
- **Attention as Logarithmic Opinion Pooling.** Relaxing the uniform reliability assumption — allowing weights to depend on query-signal interaction — yields the attention mechanism, mathematically equivalent to Hinton's Product of Experts with context-dependent reliability (Theorem 8.3). This provides a probabilistic justification for why attention computes a weighted sum.
- **Exact neural pruning.** WAND and Block-Max WAND algorithms from information retrieval constitute provably exact pruning methods for sigmoid-bounded neural structures, with formal safety guarantees unattainable with ReLU's unboundedness.

- **Depth from recursive inference.** Deep networks are chains of iterated marginalization over latent variables (Theorem 9.1.1). Each layer constructs the evidence required by the next.

3. Core Concepts

3.1 The Posting List

The posting list is the central abstraction in UQA. Every query operation — regardless of paradigm — produces a posting list.

A posting list is an ordered sequence of (doc_id, payload) pairs, sorted by doc_id. Each payload contains:

- **positions**: term positions within the document (for text retrieval)
- **score**: a relevance score (float)
- **fields**: a dictionary of metadata

The ordering by doc_id enables efficient two-pointer merge algorithms for Boolean operations:

- **Union** (OR): merge two sorted lists, keeping all entries
- **Intersection** (AND): merge two sorted lists, keeping only shared doc_ids
- **Complement** (NOT): subtract one list from the universal set

These operations form a complete Boolean algebra, meaning any Boolean combination of any paradigm's results is well-defined and efficiently computable.

3.2 Operators

Operators are functions that produce posting lists. They are the building blocks of query plans.

Primitive operators:

Operator	Input	Output	Description
TermOperator	term string	PostingList	Documents containing the term
KNNOperator	query vector, k	PostingList	k nearest neighbors by cosine similarity
VectorSimilarityOperator	query vector, θ	PostingList	Documents with similarity $\geq \theta$
FilterOperator	field, predicate	PostingList	Documents matching the predicate
ScoreOperator	source, scorer	PostingList	Adds relevance scores to documents
FacetOperator	field	dict	Value counts for a field

Boolean operators:

Operator	Semantics
UnionOperator	$A_1 \cup A_2 \cup \cdots \cup A_n$
IntersectOperator	$A_1 \cap A_2 \cap \cdots \cap A_n$
ComplementOperator	$\overline{A}$

Fusion operators:

Operator	Semantics
LogOddsFusionOperator	Combine calibrated probabilities via log-odds mean
ProbBoolFusionOperator	Probabilistic AND or OR
ProbNotOperator	Probabilistic NOT: $P' = 1 - P$

Graph operators:

Operator	Semantics
TraverseOperator	BFS traversal from a vertex
PatternMatchOperator	Subgraph isomorphism matching
RegularPathQueryOperator	NFA-based regular path evaluation

All operators implement an `execute(context)` method that returns a PostingList, and a `cost_estimate(stats)` method used by the query optimizer.

3.3 Scoring

UQA supports multiple scoring models, all of which produce values in $[0, 1]$:

BM25 (traditional lexical scoring):

$$\text{score}(f, n) = w - \frac{w}{1 + f \cdot \text{inv_norm}} \tag{1}$$

where $w = \text{boost} \cdot \text{IDF}(t)$ and inv_norm encodes document length normalization. Parameters: $k_1 = 1.2$, $b = 0.75$.

Bayesian BM25 (calibrated probabilistic scoring): Transforms BM25 raw scores into $P(\text{relevant}) \in [0, 1]$ via Bayesian inference with a sigmoid likelihood model. The posterior decomposes into three additive log-odds terms: likelihood, base rate, and document prior.

Vector similarity (semantic scoring): Cosine similarity in $[-1, 1]$ is linearly mapped to probability space: $P = (1 + \text{sim})/2$.

3.4 Multi-Signal Fusion

When combining signals from different paradigms, UQA offers two approaches:

Log-odds conjunction (recommended): Maps each calibrated probability to log-odds via the logit function, computes the arithmetic mean, scales by n^α (default $\alpha = 0.5$ for $\sqrt{n}$ confidence scaling), and applies sigmoid to return to probability space. This resolves the conjunction shrinkage problem: agreement among signals amplifies confidence without penalizing for signal count.

$$P_{\text{final}} = \sigma\left(\frac{1}{n^{1-\alpha}} \sum_{i=1}^{n} \text{logit}(P_i)\right) \tag{2}$$

Properties:

- Scale neutrality: if all $P_i = p$, then $P_{\text{final}} = p$ regardless of n
- Sign preservation: agreement among irrelevant signals cannot produce relevance
- Irrelevance preservation: all $P_i < 0.5$ implies $P_{\text{final}} < 0.5$
- Relevance preservation: all $P_i > 0.5$ implies $P_{\text{final}} > 0.5$

Probabilistic Boolean: Standard probability rules under independence.

- AND: $P = \prod_i P_i$
- OR: $P = 1 - \prod_i (1 - P_i)$
- NOT: $P = 1 - P_{\text{signal}}$

4. Function Reference

4.1 WHERE-Clause Functions

`text_match(field, query)`

Full-text search with BM25 scoring. Tokenizes the query string, retrieves the posting list for each token from the inverted index, intersects them, and scores results using BM25.

```
SELECT title, _score FROM papers
WHERE text_match(title, 'attention transformer')
ORDER BY _score DESC;
```

Parameter	Type	Description
`field`	column name	The text column to search
`query`	string	Space-separated query terms

Returns: PostingList with `_score` set to the BM25 score (≥ 0, unbounded).

`bayesian_match(field, query)`

Bayesian BM25 search. Same tokenization and retrieval as `text_match`, but applies the Bayesian calibration transform so that `_score` is $P(\text{relevant}) \in [0, 1]$.

```
SELECT title, _score FROM papers
WHERE bayesian_match(title, 'attention transformer')
ORDER BY _score DESC;
```

Parameter	Type	Description
`field`	column name	The text column to search
`query`	string	Space-separated query terms

Returns: PostingList with `_score` as a calibrated probability in $[0, 1]$.

`knn_match(k)`

K-nearest neighbor vector search using the HNSW index. Requires calling `engine.set_query_vector(vec)` before executing the SQL statement.

```
engine.set_query_vector(np.array([...], dtype=np.float32))
result = engine.sql("SELECT title, _score FROM papers WHERE knn_match(10)
ORDER BY _score DESC")
```

Parameter	Type	Description
`k`	integer	Number of nearest neighbors to retrieve

Returns: PostingList with `_score` set to cosine similarity.

`traverse_match(start, label, max_hops)`

Graph reachability as a scored signal. Performs BFS from a starting vertex along edges with the given label, up to `max_hops` hops. Reachable vertices receive a score of 0.9.

```
SELECT title, _score FROM papers
WHERE traverse_match(1, 'cited_by', 2)
ORDER BY _score DESC;
```

Parameter	Type	Description
`start`	integer	Starting vertex ID
`label`	string	Edge label to follow
`max_hops`	integer	Maximum traversal depth (default: 1)

Returns: PostingList of reachable vertices with `_score = 0.9`.

`path_filter(path, value)` / `path_filter(path, op, value)`

Filter on nested (hierarchical) document fields. The 2-argument form tests equality; the 3-argument form applies a comparison operator.

```
-- Equality
SELECT * FROM _default WHERE path_filter('shipping.city', 'Seoul');

-- Comparison
SELECT * FROM _default WHERE path_filter('items.price', '>', 100);
```

Parameter	Type	Description
`path`	string	Dot-separated path into the nested document
`op`	string	Comparison operator: `=`, `!=`, `<`, `>`, `<=`, `>=`
`value`	any	Value to compare against

If any element in a nested array matches, the document is included.

`vector_exclude(k, threshold)`

KNN search with negative vector exclusion. Retrieves k nearest neighbors to the query vector, then removes documents whose cosine similarity to the negative vector exceeds the threshold. Requires both `set_query_vector()` and `set_negative_vector()`.

```
engine.set_query_vector(query_vec)
engine.set_negative_vector(negative_vec)
result = engine.sql("SELECT title, _score FROM papers WHERE
vector_exclude(10, 0.8) ORDER BY _score DESC")
```

Parameter	Type	Description
`k`	integer	Number of nearest neighbors to retrieve
`threshold`	float	Exclusion threshold for negative similarity

4.2 Fusion Meta-Functions

Fusion functions combine multiple signal functions into a single relevance score. Each signal argument must be a signal function call (`text_match`, `bayesian_match`, `knn_match`, or `traverse_match`). Inside a fusion context, `text_match` is automatically promoted to Bayesian BM25 calibration so all signals produce probabilities in $(0, 1)$.

`fuse_log_odds(signal1, signal2, ...[, alpha])`

Log-odds conjunction. Maps each signal's calibrated probability to log-odds, computes the mean, scales by n^α, and applies sigmoid. The optional last numeric argument sets α (default: 0.5).

```
SELECT title, _score FROM papers
WHERE fuse_log_odds(
    text_match(title, 'attention'),
    knn_match(5),
    traverse_match(1, 'cited_by', 2)
) ORDER BY _score DESC;

-- Custom alpha
SELECT title, _score FROM papers
WHERE fuse_log_odds(
    bayesian_match(title, 'neural'),
    knn_match(10),
    0.7
) ORDER BY _score DESC;
```

`fuse_prob_and(signal1, signal2, ...)`

Probabilistic AND. Computes $P = \prod_i P_i$.

```
SELECT title, _score FROM papers
WHERE fuse_prob_and(
    text_match(title, 'attention'),
    knn_match(5)
) ORDER BY _score DESC;
```

`fuse_prob_or(signal1, signal2, ...)`

Probabilistic OR. Computes $P = 1 - \prod_i (1 - P_i)$.

```
SELECT title, _score FROM papers
WHERE fuse_prob_or(
    text_match(title, 'attention'),
    knn_match(5)
) ORDER BY _score DESC;
```

`fuse_prob_not(signal)`

Probabilistic NOT. Computes $P = 1 - P_{\text{signal}}$. Takes exactly one signal argument.

```
SELECT title, _score FROM papers
WHERE fuse_prob_not(text_match(title, 'noise'))
ORDER BY _score DESC;
```

4.3 SELECT Scalar Functions

`path_agg(path, func)`

Per-row aggregation over nested arrays. Navigates the dot-separated path into each document and applies the aggregation function to the array of values found.

```
SELECT path_agg('items.price', 'sum') AS total,
       path_agg('items.quantity', 'count') AS num_items
FROM _default
WHERE path_filter('shipping.city', 'Seoul');
```

Parameter	Type	Description
`path`	string	Dot-separated path to a nested array field
`func`	string	Aggregation function: `sum`, `count`, `avg`, `min`, `max`

`path_value(path)`

Access a nested field value by dot-path.

```sql
SELECT path_value('shipping.city') AS city,
       path_value('customer.name') AS name
FROM _default;
```

Parameter	Type	Description
`path`	string	Dot-separated path to a nested field

4.4 FROM-Clause Table Functions

`traverse(start, label, max_hops)`

BFS graph traversal as a virtual table. Returns one row per reachable vertex with its properties as columns.

```sql
SELECT _doc_id, _score FROM traverse(1, 'cited_by', 2);

-- With WHERE filtering on vertex properties
SELECT _doc_id, title FROM traverse(1, 'cited_by', 3)
WHERE _score > 0.5;
```

Parameter	Type	Description
`start`	integer	Starting vertex ID
`label`	string	Edge label to follow
`max_hops`	integer	Maximum traversal depth

`rpq(path_expr, start)`

Regular path query. Evaluates a regular path expression using NFA simulation and returns matching vertices.

```sql
-- Follow two 'cited_by' edges
SELECT _doc_id, title FROM rpq('cited_by/cited_by', 1);

-- Alternation: follow 'cited_by' or 'references'
SELECT _doc_id FROM rpq('cited_by|references', 1);
```

Parameter	Type	Description
`path_expr`	string	Path expression using `/` (concatenation) and `|` (alternation)
`start`	integer	Starting vertex ID (optional)

`text_search(query, field, table)`

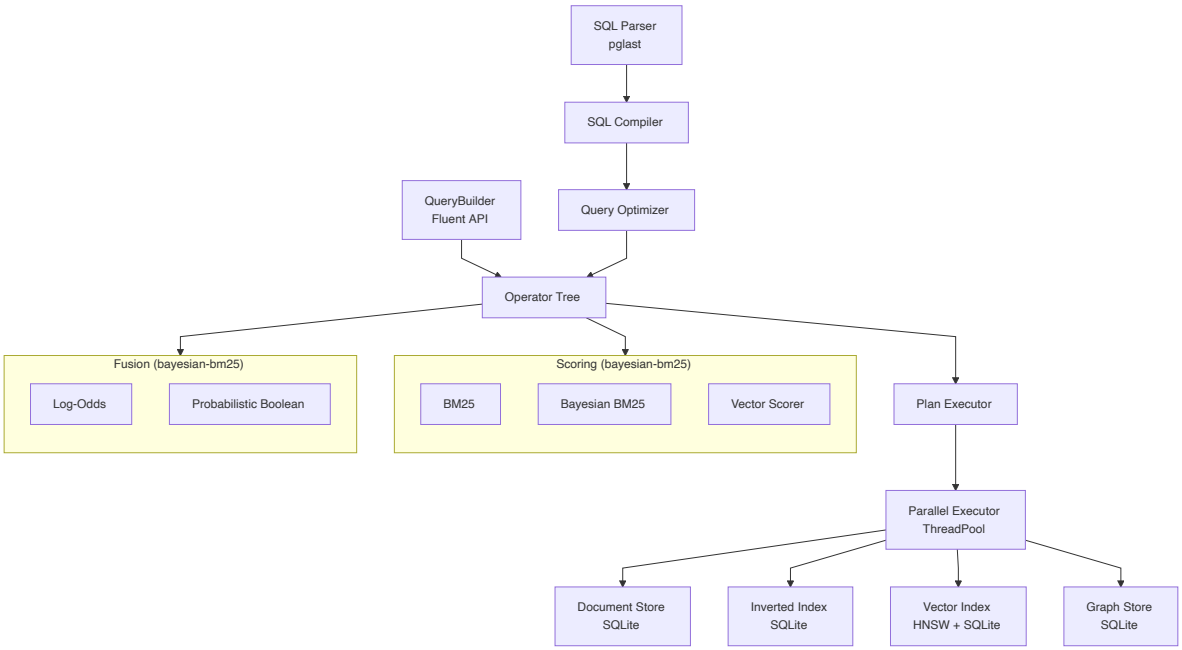
Table-scoped full-text search as a FROM-clause source. Returns documents from the specified table matching the query.

```sql
SELECT _doc_id, _score FROM text_search('attention', 'title', 'papers')
ORDER BY _score DESC;
```

Parameter	Type	Description
`query`	string	Search query
`field`	string	Field to search
`table`	string	Table name (optional; defaults to current table context)

5. Architecture

5.1 High-Level Architecture



5.2 Storage Layer

All storage backends have two implementations: an in-memory variant for ephemeral use and a SQLite-backed variant for persistence.

DocumentStore / SQLiteDocumentStore: Stores full documents. Per-table isolation: each SQL table gets its own `_data_{table_name}` SQLite table with typed columns.

InvertedIndex / SQLiteInvertedIndex: Maps (field, term) pairs to posting lists. Includes skip pointers for fast forward-seeking during intersection and block-max indexes for WAND pruning. Per-table, per-field SQLite tables: `_inverted_{table}_{field}`.

HNSWIndex / SQLiteVectorIndex: Hierarchical Navigable Small World graph for approximate nearest neighbor search. Cosine distance metric. Configuration: ef_construction=200, M=16. Vectors stored as float32 blobs in the `_vectors` table.

GraphStore / SQLiteGraphStore: Adjacency-list graph storage with indexes on (source_id, label), (target_id, label), and (label). Vertex and edge properties stored as JSON.

Catalog: SQLite-based system catalog managing metadata, table schemas, documents, graph data, vectors, postings, statistics, and scoring parameters. Write-through semantics: every mutation writes to both in-memory structures and SQLite immediately. WAL mode for concurrent reads.

5.3 SQL Compiler

The SQL compiler uses pglast (PostgreSQL parser) to parse SQL into an AST, then compiles each statement into the appropriate operation:

- **DDL**: CREATE/DROP TABLE, VIEW, INDEX; PREPARE/EXECUTE/DEALLOCATE
- **DML**: INSERT, UPDATE, DELETE with WHERE filtering
- **DQL**: SELECT with DISTINCT, WHERE, GROUP BY, HAVING, ORDER BY, LIMIT, OFFSET
- **Joins**: INNER JOIN, LEFT JOIN with ON conditions
- **Subqueries**: IN, EXISTS, scalar, correlated
- **CTEs**: WITH ... AS
- **Window functions**: ROW_NUMBER, RANK, DENSE_RANK, NTILE, LAG, LEAD
- **Transactions**: BEGIN, COMMIT, ROLLBACK, SAVEPOINT
- **Utility**: EXPLAIN, ANALYZE

The compiler translates WHERE clauses into operator trees (union/intersect/complement), recognizing extended functions like `text_match`, `knn_match`, `traverse_match`, and fusion meta-functions like `fuse_log_odds`.

5.4 Query Optimizer

The optimizer applies equivalence-preserving rewrite rules:

1. **Filter pushdown**: Pushes FilterOperator through IntersectOperator to reduce intermediate result sizes early.
2. **Graph pattern filter pushdown**: Incorporates filter predicates into vertex pattern constraints, pruning during matching rather than post-filtering.
3. **Vector threshold merge**: Combines multiple vector similarity operators with the same query vector into a single HNSW search.
4. **Intersect operand reordering**: Sorts operands by estimated cardinality (cheapest first) for optimal two-pointer intersection.
5. **Fusion signal reordering**: Sorts signals by cost estimate for early termination.
6. **Index scan substitution**: Replaces full-table FilterOperator scans with B-tree index scans when the estimated cost is lower.

The optimizer uses a **CardinalityEstimator** backed by per-column statistics: equi-depth histograms and Most Common Values (MCVs), populated by the ANALYZE command.

5.5 Execution Engine

Plan Executor: Recursively walks the operator tree, executing each operator and collecting timing statistics. The EXPLAIN command produces a tree-formatted query plan.

Parallel Executor: Independent operator branches (children of Union, Intersect, and Fusion operators) execute concurrently via ThreadPoolExecutor. Configurable via `parallel_workers` (default: 4, 0 to disable).

Volcano Iterator Engine: Pull-based pipelined execution using Apache Arrow columnar batches (RecordBatch). Physical operators implement open/next/close. Selection vectors enable lazy filtering without materializing intermediate results.

Disk Spilling: Blocking operators (sort, hash-aggregate, distinct) spill intermediate data to temporary Arrow IPC files when input exceeds `spill_threshold` rows:

Operator	Strategy
SortOp	External merge sort with k-way min-heap merge
HashAggOp	Grace hash: partition into 16 on-disk files, aggregate each independently
DistinctOp	Hash partition dedup: same 16-partition strategy

6. Usage

6.1 Installation

```
pip install uqa

# From source
pip install -e .

# With development dependencies
pip install -e ".[dev]"
```

Requirements: Python 3.12+, numpy >= 1.26, pyarrow >= 20.0, bayesian-bm25 >= 0.8.0, hnswlib >= 0.8, pglast >= 7.0.

6.2 Creating an Engine

```
from uqa.engine import Engine

# In-memory (ephemeral)
engine = Engine(vector_dimensions=64, max_elements=10000)

# Persistent (SQLite-backed)
engine = Engine(db_path="research.db", vector_dimensions=64)

# With tuning
engine = Engine(
    db_path="research.db",
    vector_dimensions=64,
    max_elements=100000,
    parallel_workers=4,     # concurrent operator execution
    spill_threshold=100000  # spill to disk after 100K rows
)

```

```python
# Context manager
with Engine(db_path="research.db") as engine:
    ...
```

6.3 SQL API

The SQL API is the primary interface for UQA.

Schema Definition

```sql
CREATE TABLE papers (
    id SERIAL PRIMARY KEY,
    title TEXT NOT NULL,
    year INTEGER NOT NULL,
    field TEXT,
    citations INTEGER DEFAULT 0
);
```

Data Manipulation

```sql
INSERT INTO papers (title, year, field, citations) VALUES
    ('attention is all you need', 2017, 'NLP', 90000),
    ('bert pre-training', 2019, 'NLP', 75000),
    ('graph neural networks', 2018, 'graph', 45000);

UPDATE papers SET citations = 95000 WHERE title = 'attention is all you need';

DELETE FROM papers WHERE citations < 1000;
```

Relational Queries

```sql
-- Standard SQL
SELECT title, year FROM papers WHERE year >= 2018 ORDER BY year DESC LIMIT 10;

-- Aggregation
SELECT field, COUNT(*) AS cnt, AVG(citations) AS avg_cites
FROM papers GROUP BY field HAVING COUNT(*) > 1;

-- Joins
SELECT p.title, a.name
FROM papers p INNER JOIN authors a ON p.id = a.paper_id;

-- Subqueries
SELECT title FROM papers
WHERE citations > (SELECT AVG(citations) FROM papers);

-- CTEs
WITH top_papers AS (
    SELECT * FROM papers WHERE citations > 50000
)
SELECT field, COUNT(*) FROM top_papers GROUP BY field;
```

```sql

-- Window functions
SELECT title, year,
    ROW_NUMBER() OVER (PARTITION BY field ORDER BY citations DESC) AS rank
FROM papers;

-- Views
CREATE VIEW recent_papers AS SELECT * FROM papers WHERE year >= 2020;
SELECT * FROM recent_papers;

-- Prepared statements
PREPARE find_by_year(int) AS SELECT title FROM papers WHERE year = $1;
EXECUTE find_by_year(2017);
DEALLOCATE find_by_year;

-- Transactions
BEGIN;
INSERT INTO papers (title, year) VALUES ('new paper', 2024);
SAVEPOINT sp1;
DELETE FROM papers WHERE year < 2010;
ROLLBACK TO SAVEPOINT sp1;
COMMIT;
```

Full-Text Search

```sql
-- BM25 scoring (raw scores)
SELECT title, _score FROM papers
WHERE text_match(title, 'attention transformer')
ORDER BY _score DESC;

-- Bayesian BM25 (calibrated probabilities)
SELECT title, _score FROM papers
WHERE bayesian_match(title, 'attention transformer')
ORDER BY _score DESC;
```

`text_match(field, query)` tokenizes the query, retrieves posting lists for each term, intersects them, and scores results using BM25. `bayesian_match` does the same but applies the Bayesian calibration transform, producing $P(\text{relevant}) \in [0, 1]$.

Vector Search

```python
import numpy as np

# Set the query vector before executing SQL
engine.set_query_vector(np.random.randn(64).astype(np.float32))
```

```sql
-- K-nearest neighbors
SELECT title, _score FROM papers WHERE knn_match(10) ORDER BY _score DESC;
```

Graph Queries

```sql
-- Traverse: BFS from vertex 1 along 'cited_by' edges, up to 2 hops
SELECT _doc_id, title FROM traverse(1, 'cited_by', 2);

-- Regular path query: follow two 'cited_by' edges
SELECT _doc_id, title FROM rpq('cited_by/cited_by', 1);

-- Graph traversal as a scored signal in WHERE
SELECT title, _score FROM papers
WHERE traverse_match(1, 'cited_by', 2) ORDER BY _score DESC;
```

Multi-Signal Fusion

```sql
-- Log-odds fusion of text, vector, and graph signals
SELECT title, _score FROM papers
WHERE fuse_log_odds(
    text_match(title, 'attention'),
    knn_match(5),
    traverse_match(1, 'cited_by', 2)
) ORDER BY _score DESC;

-- Custom confidence scaling (alpha = 0.7)
SELECT title, _score FROM papers
WHERE fuse_log_odds(
    bayesian_match(title, 'neural'),
    knn_match(10),
    0.7
) ORDER BY _score DESC;

-- Probabilistic Boolean
SELECT title, _score FROM papers
WHERE fuse_prob_and(
    text_match(title, 'attention'),
    knn_match(5)
) ORDER BY _score DESC;

SELECT title, _score FROM papers
WHERE fuse_prob_or(
    text_match(title, 'attention'),
    knn_match(5)
) ORDER BY _score DESC;
```

Hierarchical Data

```sql
-- Filter on nested paths
SELECT * FROM _default WHERE path_filter('shipping.city', 'Seoul');

-- Comparison operators on nested paths
SELECT * FROM _default WHERE path_filter('items.price', '>', 100);

-- Aggregate nested arrays
SELECT path_agg('items.price', 'sum') AS total FROM _default
WHERE path_filter('shipping.city', 'Seoul');

-- Access nested field values
SELECT path_value('shipping.city') AS city FROM _default;
```

Query Introspection

```sql
-- Show the query execution plan
EXPLAIN SELECT title, _score FROM papers
WHERE text_match(title, 'attention') ORDER BY _score DESC;

-- Collect column statistics for the optimizer
ANALYZE papers;
```

6.4 QueryBuilder Fluent API

The fluent API provides a programmatic alternative to SQL.

```python
from uqa.engine import Engine
from uqa.core.types import Equals, GreaterThanOrEqual
import numpy as np

engine = Engine(vector_dimensions=64)

# Term search
result = engine.query().term("attention", field="title").execute()

# BM25 scoring
result = (
    engine.query()
    .term("attention", field="title")
    .score_bm25("attention")
    .execute()
)

# Bayesian BM25 scoring
result = (
    engine.query()
    .term("attention", field="title")
    .score_bayesian_bm25("attention")
    .execute()
)

# Boolean composition
relevant = engine.query().term("attention")
```

```python
not_graph = engine.query().term("graph").not_()
result = relevant.and_(not_graph).execute()

# KNN vector search
query_vec = np.random.randn(64).astype(np.float32)
result = engine.query().knn(query_vec, k=10).execute()

# Hybrid: text AND vector
text = engine.query().term("transformer").score_bm25("transformer")
vector = engine.query().knn(query_vec, k=10)
result = text.and_(vector).execute()

# Filter
result = (
    engine.query()
    .term("attention")
    .filter("year", GreaterThanOrEqual(2020))
    .execute()
)

# Facets
facets = engine.query().facet("field")  # dict[value, count]

# Aggregation
count = (
    engine.query()
    .filter("year", GreaterThanOrEqual(2020))
    .aggregate("id", "count")
)

# Graph traversal
team = engine.query().traverse(start_id=1, label="manages", max_hops=2)
members = team.execute()
total_salary = team.vertex_aggregate("salary", "sum")

# Join
result = (
    engine.query().term("attention")
    .join(
        engine.query().term("transformer"),
        left_field="id",
        right_field="paper_id"
    )
    .execute()
)

# Explain
plan = engine.query().term("attention").explain()
print(plan)
```

6.5 Programmatic Document API

For applications that do not use SQL tables, the Engine provides a direct document API with a global (table-independent) document store.

```python
import numpy as np
from uqa.engine import Engine
from uqa.graph.store import Vertex, Edge

engine = Engine(db_path="data.db", vector_dimensions=64)

# Add documents with embeddings
engine.add_document(
    doc_id=1,
    document={"title": "attention is all you need", "year": 2017},
    embedding=np.random.randn(64).astype(np.float32)
)

# Delete documents
engine.delete_document(doc_id=1)

# Add graph vertices and edges
engine.add_graph_vertex(Vertex(vertex_id=1, properties={"name": "Alice"}))
engine.add_graph_vertex(Vertex(vertex_id=2, properties={"name": "Bob"}))
engine.add_graph_edge(Edge(
    edge_id=1,
    source_id=1,
    target_id=2,
    label="manages",
    properties={"since": 2020}
))

# Persist scoring parameters for Bayesian BM25 calibration
engine.save_scoring_params("text_signal", {"alpha": 1.0, "beta": 3.5})
params = engine.load_scoring_params("text_signal")
```

6.6 Interactive SQL Shell

UQA includes an interactive SQL shell with syntax highlighting and autocompletion.

```python
python usql.py                  # In-memory
python usql.py --db mydata.db   # Persistent database
```

Shell commands:

Command	Description
`\dt`	List tables
`\d <table>`	Describe table schema
`\ds <table>`	Show column statistics (requires ANALYZE first)
`\di`	List indexes
`\timing`	Toggle query timing display
`\reset`	Reset the engine
`\q`	Quit

6.7 Transactions

UQA supports ACID transactions with savepoints through the SQL interface and the programmatic API.

```
# SQL transactions
engine.sql("BEGIN")
engine.sql("INSERT INTO papers (title, year) VALUES ('new paper', 2024)")
engine.sql("SAVEPOINT sp1")
engine.sql("DELETE FROM papers WHERE year < 2010")
engine.sql("ROLLBACK TO SAVEPOINT sp1")  # undo the DELETE
engine.sql("COMMIT")

# Programmatic transactions
txn = engine.begin()
try:
    engine.add_document(1, {"title": "test"})
    txn.commit()
except Exception:
    txn.rollback()
```

7. Significance

7.1 Unification Through Algebra

The central contribution of UQA is demonstrating that four historically separate data paradigms can be unified under a single algebraic structure without sacrificing the expressivity of any individual paradigm. The posting list abstraction is not a lowest-common-denominator compromise but a genuine isomorphism: every operation in the Boolean algebra of posting lists has a well-defined semantics in each paradigm, and cross-paradigm compositions inherit the algebraic properties (commutativity, associativity, distributivity) of the underlying Boolean algebra.

This means, for example, that a query combining BM25 text scoring with KNN vector search and graph traversal is not an ad-hoc pipeline but a composition of operators whose algebraic properties are formally proven. The query optimizer can apply rewriting rules (filter pushdown, operand reordering) with guaranteed semantic preservation because the rules are derived from

the algebraic structure, not from heuristics.

7.2 Closing the Probabilistic Relevance Gap

For nearly fifty years, BM25 scores — despite originating from a probabilistic model — have been treated as opaque numbers rather than probabilities. Bayesian BM25 closes this gap by returning BM25 to the probability space from which it was derived. The practical consequence is that BM25 scores become directly comparable with and combinable with other probabilistic signals (vector similarity, graph reachability) without ad-hoc normalization.

The log-odds conjunction framework resolves a fundamental problem with naive probabilistic combination: the conjunction shrinkage problem, where multiplying independent probability estimates drives the combined score toward zero as more signals agree. The log-odds mean with $\sqrt{n}$ confidence scaling preserves the direction of evidence while amplifying its magnitude proportionally to the number of agreeing signals.

7.3 Neural Structure from Probability

The fourth paper reveals that the multi-signal fusion pipeline — calibrate, transform to log-odds, aggregate, apply sigmoid — has the exact structure of a feedforward neural network. This is not an analogy; the correspondence is proven algebraically. The result reverses the conventional direction of explanation in neural network theory: rather than designing a neural architecture and analyzing it probabilistically, UQA begins with probability and arrives at neural structure as a mathematical consequence.

This has implications beyond information retrieval. The derivation provides probabilistic origins for the dominant activation functions (sigmoid, ReLU, Swish, GELU), shows that attention is Logarithmic Opinion Pooling with context-dependent reliability, and establishes that depth arises from iterated marginalization over latent variables.

8. Examples

Example scripts are provided in the `examples/` directory:

Fluent API (`examples/fluent/`)

Script	Topics
`text_search.py`	BM25, Bayesian BM25, boolean composition, facets, aggregation
`vector_and_hybrid.py`	KNN, hybrid search, vector exclusion, log-odds fusion
`graph.py`	Traversal, RPQ, pattern matching, graph indexes
`hierarchical.py`	Nested data, path filters, path aggregation

SQL (`examples/sql/`)

Script	Topics
`basics.py`	DDL, DML, SELECT, CTE, window functions, transactions, views
`functions.py`	text_match, knn_match, path_agg, path_value, path_filter
`graph.py`	FROM traverse/rpq, aggregates, GROUP BY, WHERE
`fusion.py`	fuse_log_odds, fuse_prob_and/or/not, EXPLAIN

9. Testing

Tests use pytest with hypothesis for property-based testing.

```
# Run all tests
python -m pytest uqa/tests/ -v

# Run a specific test file
python -m pytest uqa/tests/test_sql.py -v
```

Key test areas:

Test file	Coverage
`test_posting_list.py`	Core PostingList algebra
`test_operators.py`	All operator types
`test_scoring.py`	BM25, Bayesian BM25, vector scoring
`test_fusion.py`	Log-odds, probabilistic boolean fusion
`test_sql.py`	SQL compilation and execution
`test_execution.py`	Physical operators, Volcano iterator
`test_joins.py`	All join types
`test_graph.py`	Graph store, traversal, pattern matching, RPQ
`test_index.py`	Inverted index, skip pointers, block-max
`test_sqlite_*.py`	SQLite backend persistence
`test_transaction.py`	BEGIN, COMMIT, ROLLBACK, SAVEPOINT
`test_cost_optimizer.py`	Cardinality estimation, rewrite rules
`test_parallel.py`	Parallel execution correctness
`test_integration.py`	End-to-end scenarios

10. References

1. Jeong, J. (2023). A Unified Mathematical Framework for Query Algebras Across

Heterogeneous Data Paradigms.

2. Jeong, J. (2024). Extending the Unified Mathematical Framework to Support Graph Data Structures.
3. Jeong, J. (2026). Bayesian BM25: A Probabilistic Framework for Hybrid Text and Vector Search.
4. Jeong, J. (2026). From Bayesian Inference to Neural Computation: The Analytical Emergence of Neural Network Structure from Probabilistic Relevance Estimation.